

Round all answers to three decimal places when needed.

Directions: Determine if the events are independent or dependent, then find the probabilities.

- 1) Roll a 7 on an 8-sided die and a 1 on a 6-sided die.
- 2) Flip tails on a fair coin and pull a king from a standard deck.
- 3) You pull a black card, keep it, and then pull another black card.
- 4) You do not roll a six on a 6-sided die and you pull a heart from a standard deck.
- 5) You pick to play shortstop on the baseball field, and then your friend picks an outfield position.
- 6) 52 students attend a magic show. The magician gives each student a different card from a standard deck, then he has his own standard deck from which he pulls cards to call on participants. Each time the magician pulls a card, he does not replace it at any point during the show. Find the probabilities for each act.
 - a) Act 1: He pulls a King.
 - b) Act 2: He pulls a Six.
 - c) Act 3: He pulls a diamond.
 - i. Assuming the first two acts were not diamonds.
 - ii. Assuming the first two acts were diamonds.
 - d) Act 5: He pulls a red card.
 - i. Assuming Act 1 and Act 3 were both red cards.
 - ii. Assuming Act 3 was the only red card.

7) All students attending a homecoming assembly complete a digital survey as they arrive at their seats. The survey questioned them about what they were most looking forward to for the homecoming events. Assume all students in attendance participated and remained at the assembly for its entirety. Write probabilities as decimals rounded to hundredths.

-----	Bonfire	Football Game	Dance	Total
Freshmen	200	98	202	500
Sophomores	125	184	231	540
Juniors	81	294	123	498
Seniors	92	198	222	512
Total	498	774	778	2050

Teachers take turns randomly choosing students to participate in the pep assembly games. Students can get selected for multiple games.

a) What is the probability a teacher chooses a sophomore and then another sophomore for the first assembly game?

b) What is the probability a teacher chooses a student who prefers the bonfire for the second game and then a student who prefers the dance for the third game?

c) What is the probability of choosing four students for the last game who are all looking forward to the football game?

8) A simple random sample of adults in a metropolitan area was selected and a survey was administered to determine political views. The results are recorded below:

Political Views				
Age	Conservative	Moderate	Liberal	Total
18-29	10	15	30	55
30-44	20	30	35	85
45-59	35	15	20	70
Over 60	20	15	10	45
Total	85	75	95	255

a) What is the probability that a person chosen at random from this sample will have moderate political views?

b) What is the probability that a person chosen at random from those in the sample who are between the ages of 30 to 44 will have moderate political views?

c) Based on your answers to (a) and (b), are political views and age independent for the population of adults in this metropolitan area? Why or why not?

9) Given that $P(A) = 0.6$, $P(B) = 0.3$, and $P(B|A) = 0.5$

a) Find $P(A \text{ and } B)$

b) Find $P(A \text{ or } B)$

c) Are events A and B independent?

10) Here are the counts (in thousands) of earned degrees in the United States in a recent year, classified by level and by gender of the degree recipient:

	Associate's	Bachelor's	Master's	Doctorate	TOTAL
Female	616	194	30	16	856
Male	529	171	44	26	770
Total	1145	365	74	42	1626

a) If you choose a degree recipient at random, what is the probability that the person you choose is female?

b) What is the conditional probability that you choose a female, given that the person chosen received a master's degree?

c) What is the conditional probability that the person chosen received a bachelor's degree, given that the person is a male?

d) Are the events "female" and "master's" independent?

11) At Portillo's in Rockford, the distribution of ages and what shift they work on a Saturday is below. Fill in the table with the missing information.

Age	Open - 1pm	1pm - 7pm	7pm - Close	Total
Under 18	6		21	29
18 to 49	4	5		11
40 and over	10	5		40
Total		12		80

Are the events "Open - 1pm" and "40 and over" independent? Show your work.